ECI 149 HW 3

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Problem 1:

1.2.2.1)

$\int_0^z \rho_o\, e^{[-z/h]} dz \times A = A \times (\rho_o \left(h - he^{(-z/h)}\right) = A \times \rho_o \times h$ (when z>>h)

1.2.2.2.1)

$\frac{n}{V} = \frac{P}{RT} = \frac{984 \times 10^2 Pa}{8.314 J/mol-air/K \times 288K} \times 28.96g - air/mol - air = 1190.12g - air/m^3$

MW of air taken from engineeringtoolbox.com

1.2.2.2.2)

$m = h \times \rho_o \times A = 7.4km(1190.21g - air/m^3)(511191863.5km^2) \times 1000^3 m^3/km^3 \times 1kg/1000g = 4.5 \times 10^{18} kg$

Assuming the radius of the earth is 6378.14 km (engineeringtoolbox.com).

1.2.2.3)

h=0.3 km

Density of seasalt aerosol, 10 micrograms per cubic meter.

$m = 0.7 \times 0.5km(10\mu g/m^3)(511191863.5km^2)(1000m/km)^3(1kg/10^9\mu g) = 1.79 \times 10^9 kg$

Problem 2:

i) According to the textbook, it is because of "geostrophic flow driven by latitudinal heating gradient[s]". East-West takes 2 weeks, North to South (same hemisphere) takes 1-2 months, according to the textbook.

ii) The intertropical convergence zone makes transport across the equator difficult, it takes around one year according to the text book.

Problem 3:

Assuming radius of the eath is 4500 km,

$V = \frac{2\pi r}{2weeks}(1000m/km)(week/7d)(d/24hr)(hr/60min)(min/60s)P = 23.4m/s$

Problem 4:

Potential temperature (θ) is the temperature that an air parcel would assume if brought adiabatically to

1000 hPa.

$\theta = T\left(\frac{P_o}{P}\right)^k$ where k=Poisson's constant = 2/7.

$\therefore T(actual\ temp) = \theta\left(\frac{P_o}{P}\right)^{-k} = \theta\left(\frac{P_o}{P_o e^{-z/h}}\right)^{-k} = \theta\left(\frac{1}{e^{-z/7.4km}}\right)^{-2/7}$

Figure 1 shows the plotted data. MATLAB was used to plot and the code is attached.

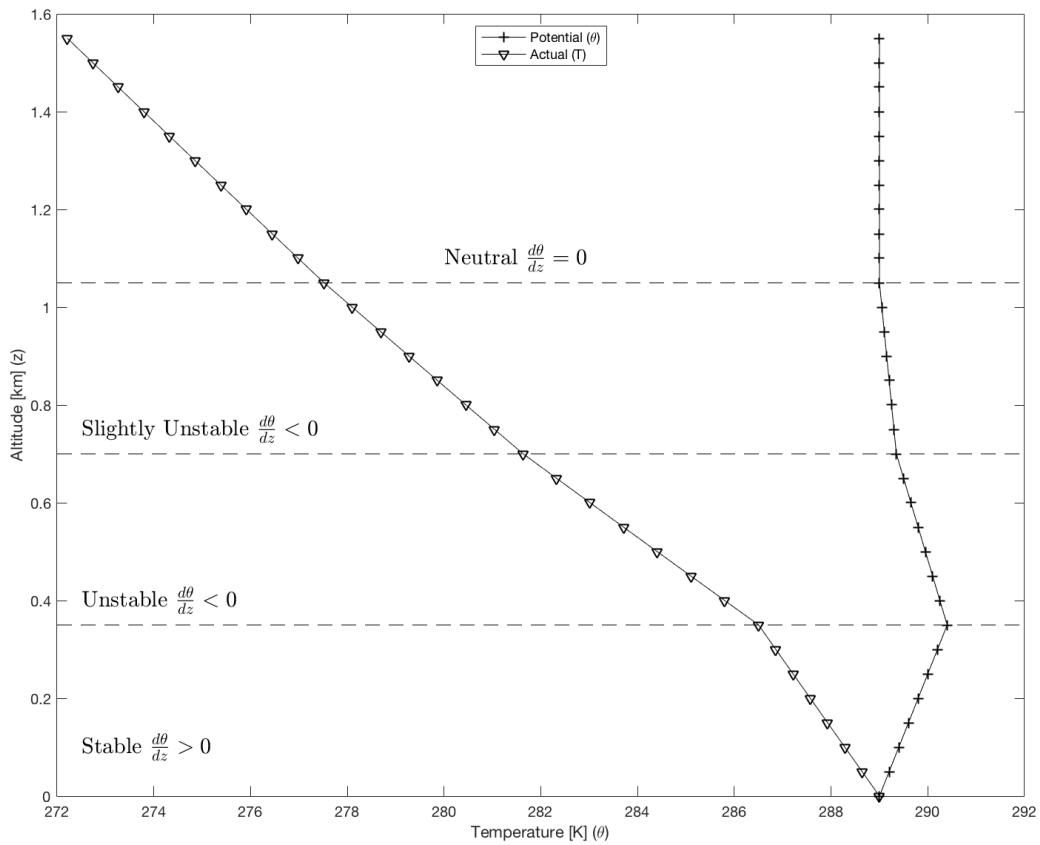


Figure 1: Actual temperature and potential temperature with stability ratings.

Problem 5:

4.4.1)

The mixing layer peaks near the end of the day because the heat emitted from the earth heats the air and results in convection cells that cause mixing. At night this effect is not as strong.

There is a mixed layer at night because the earth still emits some heat, but also because there is wind and associated turbulence that causes mixing.

4.4.2)

Figure 2 shows the concentration variations in each domain given a constant emission source from the surface. Initially the mixing layer will have an increasing concentration of X, but when the layer expands and mixes, X will be diluted. Since there is no vertical transport across the boundaries, [X] will be zero in the upper domain until the mixing layer recedes and leaves behind the dispersed pollutants.

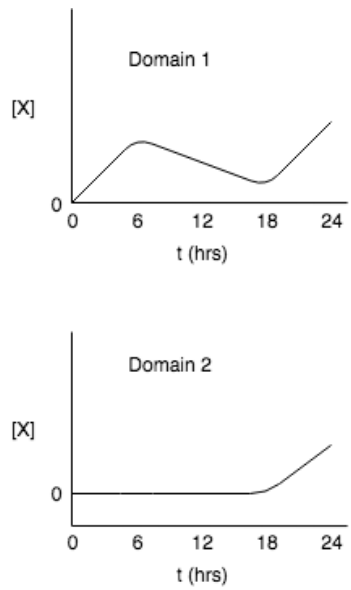


Figure 2: 4.4.2 concentration variations with constant emission at surface.

Problem 6:

Assume scale height is 7.4 km

HI: T=75F = 297.2K, z=3.4km, [NO_2]=331.25 ppb

First find the number density of air,

$\left(\frac{n}{V}\right)_{air,HI} = \frac{P}{KT} = \frac{101320Pa\ e^{-3.4/7.4}}{1.381\times10^{-23}J/molec/K\times297.2K} = 1.56\times10^{19}molec/cm^3$

And,

$\left(\frac{n}{V}\right)_{N_2O,HI} = \frac{331.25}{10^9}\times\left(\frac{n}{V}\right)_{air,HI} = 5.17\times10^{12}molec/cm^3$

ANT: T=28F=270.9K, z=0.015km, [NO_2]=330.28 ppb

First find the number density of air,

$\left(\frac{n}{V}\right)_{air,ANT} = \frac{P}{KT} = \frac{101320Pa\ e^{-0.015/7.4}}{1.381\times10^{-23}J/molec/K\times270.9K} = 2.7\times10^{19}molec/cm^3$

And,

$\left(\frac{n}{V}\right)_{N_2O,ANT} = \frac{330.28}{10^9}\times\left(\frac{n}{V}\right)_{air,ANT} = 8.92\times10^{12}molec/cm^3$

Problem 7:

i) lifetime for N_2O is longer, we can infer this because the reduction in atmospheric levels from HI to ANT are much much lower for N_2O than for CH4 (assuming no inputs along the way).

ii) It suggests that the methane source strength is stronger in the north than in the south, because the concentrations are higher there.

Problem 8:

Inversions happen when temperatures are higher at higher altitudes than at lower altitudes. This situation makes the atmosphere very stable. These events are more prone to occur at night and happen very near the surface because the earth cools most strongly from the surface.

Additionally, as air parcels sink they compress and the top of the parcel becomes warmer than the bottom, resulting in an inversion. Inversions are stronger when there are fewer clouds in the sky.

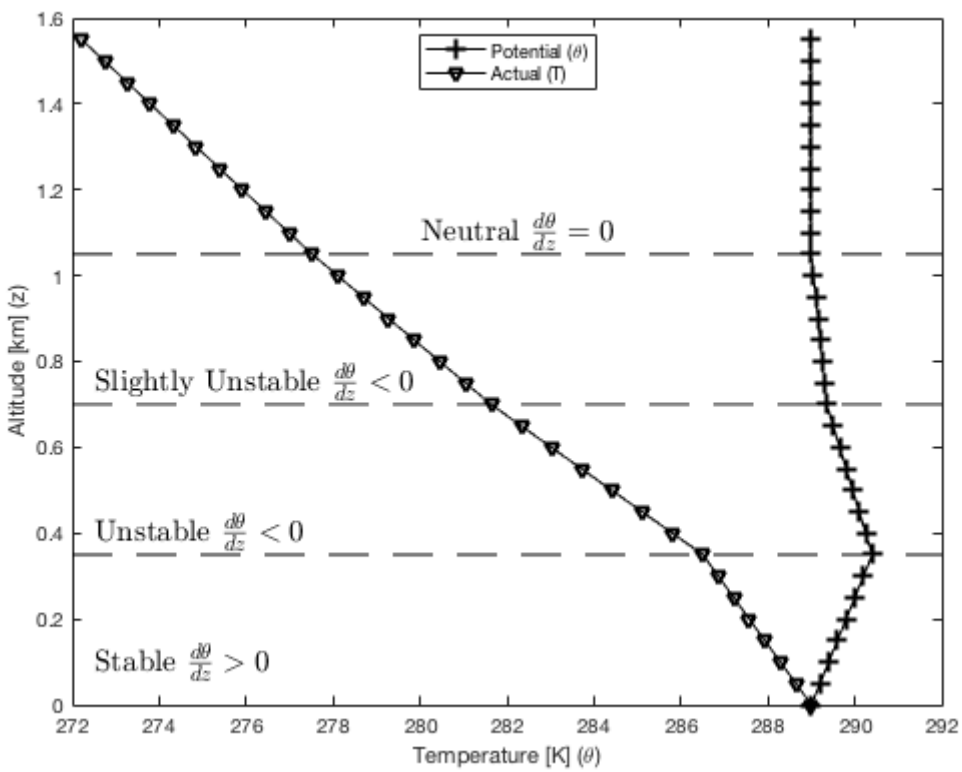
```
% ECI 149 HW3
% Charles Hammond
```

Problem 4

```
clc,clear

data=readtable('HW#3-Profile.txt');

potential_temp=data{:,2};

alt=data{:,1};
plot(potential_temp,alt,'k+-')
hold on
actual_temp=potential_temp.*(1./exp(-alt/7.4)).^(-2/7);

plot(actual_temp,alt,'kv-')

xlabel('Temperature [K] (\theta)')
ylabel('Altitude [km] (z)')

t3=text([272.5],[0.1],'Stable $\frac{d\theta}{dz}>
 0$','interpreter','latex');

plot([272 292],[.35 .35],'k--'); t2=text([272.5],[.4],'Unstable $
\frac{d\theta}{dz}< 0$','interpreter','latex')

plot([272 292],[.7 .7],'k--'); t1=text([272.5],[.75],'Slightly
 Unstable $\frac{d\theta}{dz}< 0$','interpreter','latex')

plot([272 292],[1.05 1.05],'k--'); t=text([280],[1.1],'Neutral $
\frac{d\theta}{dz} = 0$','interpreter','latex');

t.FontSize = 15;t1.FontSize = 15;t2.FontSize = 15;t3.FontSize = 15;




legend('Potential (\theta)','Actual (T)','location','North')
```

```
Warning: Variable names were modified to make them valid MATLAB
 identifiers. The
original names are saved in the VariableDescriptions property.
t2 =
  Text (Unstable $\frac{d\theta}{dz}< 0$) with properties:

                String: 'Unstable $\frac{d\theta}{dz}< 0$'
              FontSize: 10
            FontWeight: 'normal'
              FontName: 'Helvetica'
                 Color: [0 0 0]
```

```
    HorizontalAlignment: 'left'
                Position: [272.5000 0.4000 0]
                   Units: 'data'

  Use GET to show all properties
t1 =
  Text (Slightly Unstable $\frac{d\theta}{dz}< 0$) with properties:

                  String: 'Slightly Unstable $\frac{d\theta}{dz}< 0$'
                FontSize: 10
              FontWeight: 'normal'
                FontName: 'Helvetica'
                   Color: [0 0 0]
    HorizontalAlignment: 'left'
                Position: [272.5000 0.7500 0]
                   Units: 'data'

  Use GET to show all properties
```



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